

EXP NO: 2

SUPPORT VECTOR MACHINE (SVM) AND RANDOM FOREST FOR BINARY
& MULTICLASS CLASSIFICATION

AIM

To build classification models using Support Vector Machines (SVM) and Random Forest, apply them to a dataset, and evaluate the models using performance metrics like accuracy and confusion matrix.

ALGORITHM

Part A: SVM Model

1. Import necessary libraries
2. Load and explore the dataset
3. Handle missing values if any
4. Encode categorical variables
5. Split dataset into training and testing sets
6. Build SVM classifier using SVC()
7. Train and predict
8. Evaluate the model using accuracy and confusion matrix

Part B: Random Forest Model

1. Initialize Random Forest using RandomForestClassifier()
2. Train and predict
3. Evaluate and compare with SVM

CODE:

```
import pandas as pd from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split from
sklearn.preprocessing import StandardScaler from sklearn.svm
import SVC from sklearn.ensemble import
RandomForestClassifier from sklearn.metrics import
accuracy_score, confusion_matrix import seaborn as sns
import matplotlib.pyplot as plt

iris = load_iris()
X = iris.data y =
iris.target

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
X_scaled, y, test_size=0.3, random_state=42
)

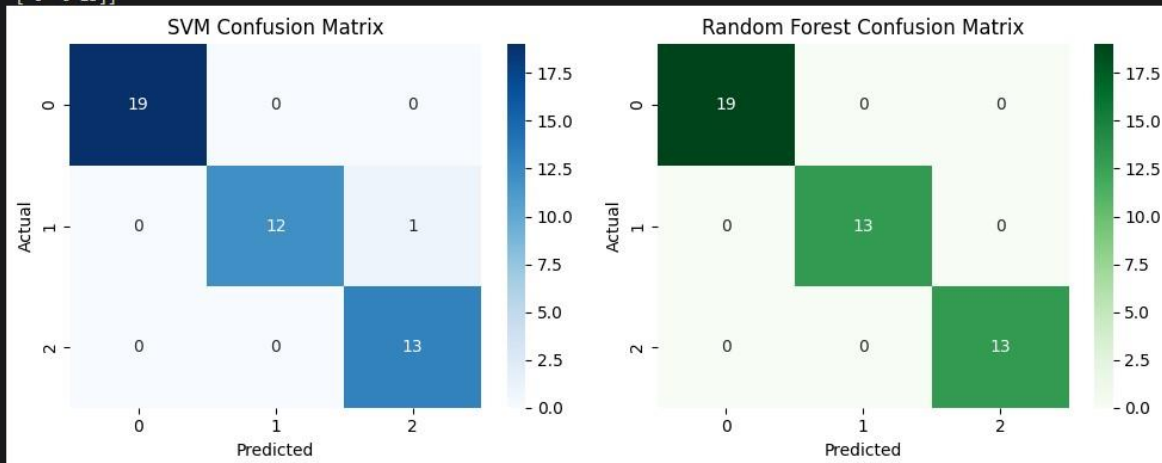
svm_model = SVC(kernel='linear') svm_model.fit(X_train,
y_train) y_pred_svm = svm_model.predict(X_test)
print("SVM Accuracy:", accuracy_score(y_test, y_pred_svm))
print("SVM Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_svm)) rf_model =
RandomForestClassifier( n_estimators=100,
random_state=42
)
rf_model.fit(X_train, y_train)
y_pred_rf = rf_model.predict(X_test) print("\nRandom Forest
Accuracy:", accuracy_score(y_test, y_pred_rf)) print("Random Forest
Confusion Matrix:") print(confusion_matrix(y_test, y_pred_rf))
```

```
plt.figure(figsize=(10, 4)) plt.subplot(1,
2, 1)
sns.heatmap( confusion_matrix(y_test,
y_pred_svm), annot=True,
cmap='Blues', fmt='d'
)
plt.title("SVM Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual") plt.subplot(1,
2, 2)
sns.heatmap(
confusion_matrix(y_test, y_pred_rf),
annot=True, cmap='Greens', fmt='d'
)
plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted") plt.ylabel("Actual")
plt.tight_layout() plt.show()
```

OUTPUT:

```
SVM Accuracy: 0.9777777777777777
SVM Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]
```

```
Random Forest Accuracy: 1.0
Random Forest Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```



RESULT:

The Support Vector Machine (SVM) and Random Forest algorithms were successfully implemented for both binary and multiclass classification tasks. The models were trained and tested on the given dataset, and both achieved good accuracy.